

5.5.4

Complete Statistics

Definition 5.5.3 (Complete Statistic)

Suppose that $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} f(x; \theta)$.

A statistic $S = s(\bar{X})$ is **complete** for the distribution if, for any function g such that $E[g(S)] = 0$ for all θ in the parameter space, we find that $g(S)$ must be 0 with probability 1.

Comments on Completeness

Completeness is useful for showing that estimators are unique in some way.

Directly showing that a statistic is complete for a distribution is often complex, so it is rarely done that way.

Examples 5.5.5 and 5.5.6 demonstrate cases where the completeness of a statistic can be shown for certain distributions.

We will focus on using a powerful theorem related to exponential families.

5.5.6

Exponential Families and Completeness

Definition 5.5.4 (One-parameter exponential family)

$X_1, X_2, \dots, X_n$ is a random sample from a **one-parameter exponential family** if θ is one-dimensional and if $X_1, X_2, \dots, X_n$ has a joint pdf that can be written as

$$f(\vec{x}; \theta) = a(\theta)b(\vec{x}) \exp(c(\theta)d(\vec{x})),$$

for some functions $a(\theta)$, $b(\vec{x})$, $c(\theta)$, and $d(\vec{x})$.

- a and c depend only on θ and not on $\vec{x}$.
- b and d depend only on $\vec{x}$ and not on θ .

Theorem 5.5.1 (Completeness and the exponential family)

If $X_1, X_2, \dots, X_n$ is a random sample from a one-parameter exponential family distribution, then

$$S = d(\vec{X})$$

is complete and sufficient for the distribution [as long as the parameter space for θ contains an open set].

Example 5.5.10

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$. Find a complete and sufficient statistic for this distribution.

Example 5.5.10 (cont)

Example 5.5.10 (cont)

Definition 5.5.5 (k -parameter exponential family)

$X_1, X_2, \dots, X_n$ is a random sample from a k -parameter exponential family if θ is k -dimensional and if $X_1, X_2, \dots, X_n$ has a joint pdf that can be written as

$$f(\vec{x}; \theta) = a(\theta)b(\vec{x}) \exp \left[\sum_{i=1}^k c_i(\theta)d_i(\vec{x}) \right]$$

for some functions $a(\theta)$, $b(\vec{x})$, $c_1(\theta), \dots, c_k(\theta)$, and $d_1(\vec{x}), \dots, d_k(\vec{x})$.

Theorem 5.5.2

If $X_1, X_2, \dots, X_n$ is a random sample from a k -parameter exponential family distribution, then

$$S = (d_1(\vec{X}), \dots, d_k(\vec{X}))$$

is complete and sufficient for the distribution [as long as the parameter space for θ contains an open set].

- E.g., a $N(\theta, \theta^2)$ distribution does not satisfy this condition.

Example 5.5.11

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Find the complete, sufficient statistics for this family of distributions.

Example 5.5.11 (cont)

Example 5.5.11 (cont)

Example 5.5.11 (cont)

5.5.5

The UMVUE

Lefhmann-Scheffé Theorem

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} f(x; \theta)$, and suppose that $S = s(\bar{X})$ is a complete and sufficient statistic for this distribution.

If $g(S)$ is some function of S that is unbiased for $\tau(\theta)$, then $g(S)$ is the UMVUE for $\tau(\theta)$.

In short: if we can find a unbiased estimator that a function of a complete sufficient statistic, then we have found the UMVUE!

Example 5.5.7

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$. Find the UMVUE for λ .

Example 5.5.7 (cont)

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Example 5.5.8

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$. Find the UMVUE for λ^2 .

Example 5.5.8 (cont)

Example 5.5.8 (cont)

Example 5.5.9

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$. Find the UMVUE for $e^{-\lambda}$.

Example 5.5.9 (cont)

Example 5.5.9 (cont)

Example 5.5.11 (cont)

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Determine the UMVUEs for μ and σ^2 .

Example 5.5.11 (cont)

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